



Módulo III

Especialización en IA Generativa y Machine Learning OPS

Herramientas para la Inteligencia Artificial
Generativa



Facultad de
Ingeniería Económica,
Estadística y Ciencias
Sociales

01

Observabilidad en
Aplicaciones GenAI

02

Evaluar
Aplicaciones GenAI

03

Gestión de prompts



LangSmith



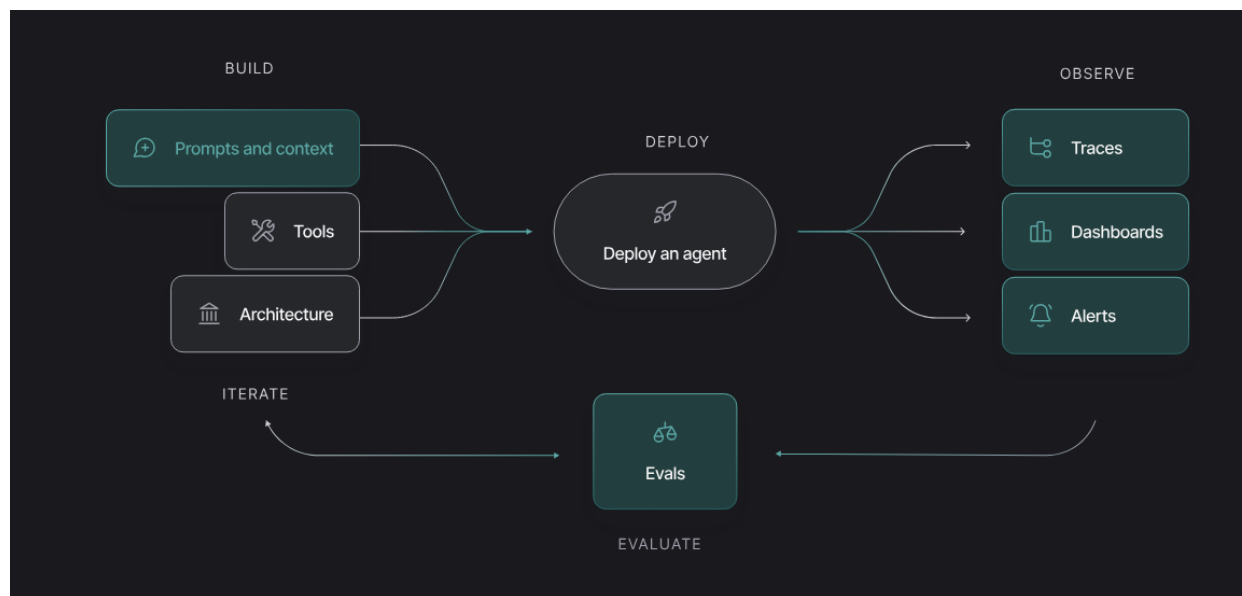
01

Observabilidad en
Aplicaciones GenAI

¿Qué es LangSmith?



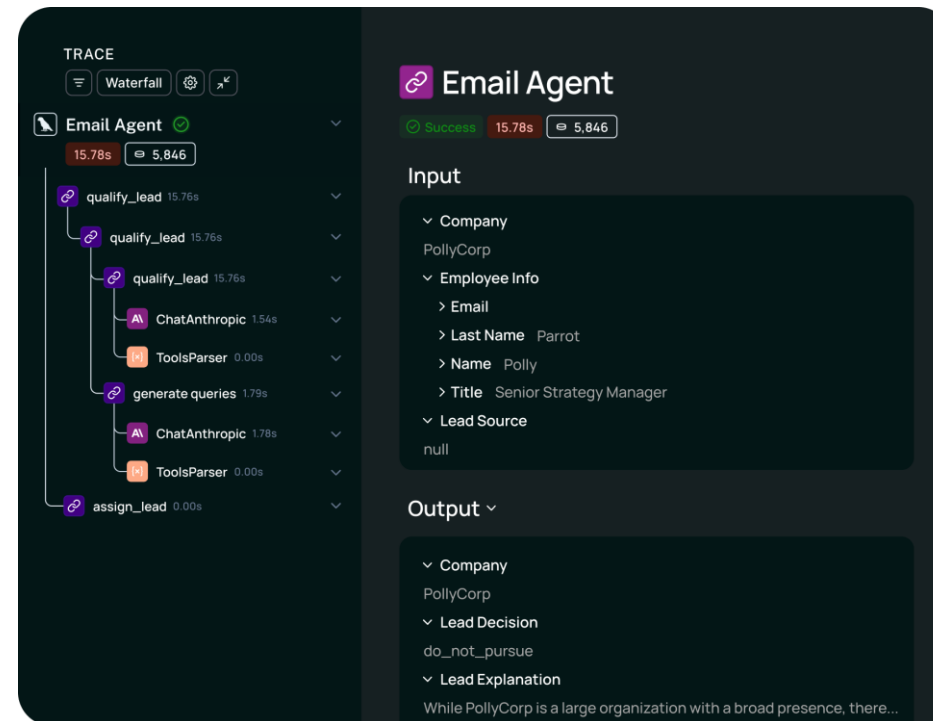
LangSmith es una plataforma unificada e independiente que proporciona herramientas esenciales para **desarrollar**, **depurar** y **gestionar** el ciclo de vida completo de las aplicaciones LLM, incluyendo el rastreo y la evaluación integrados en producción.



La plataforma de LangSmith



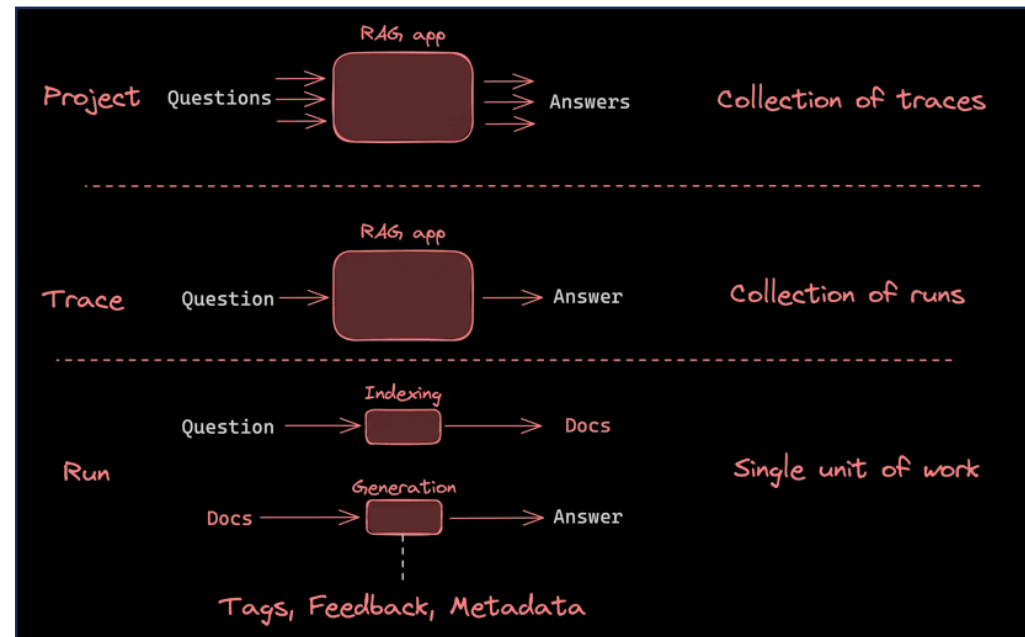
Es un **entorno de desarrollo** (IDE) especializado en agentes que permite la **visualización, interacción y depuración** de sistemas basados en el protocolo Agent Server API, e integra funcionalidades clave como el trazado (tracing), la evaluación y la ingeniería de prompts.



Observabilidad



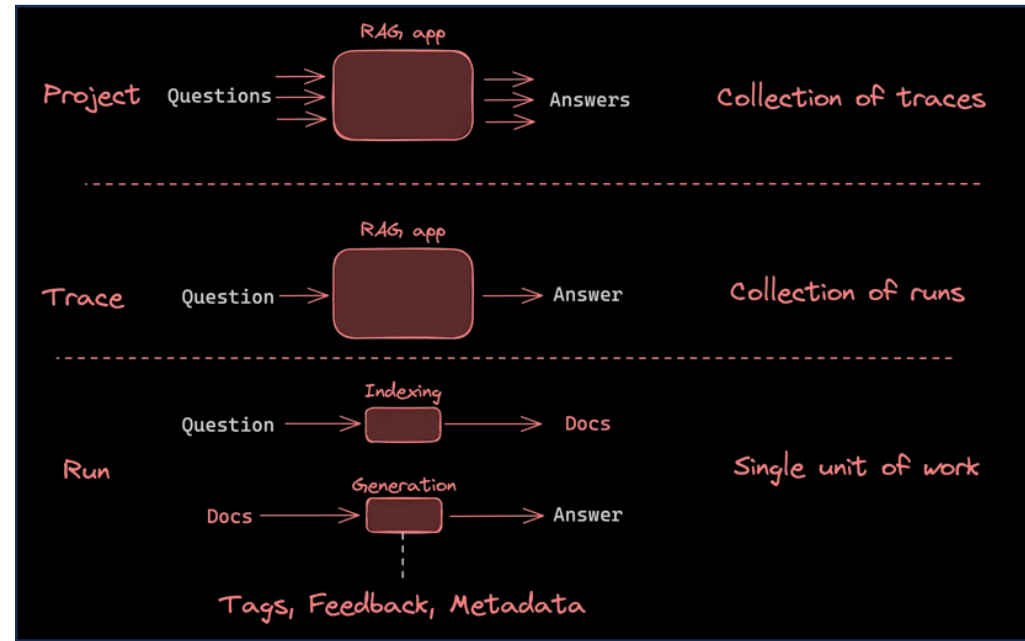
Las aplicaciones que utilizan LLMs (Large Language Models) pueden generar diferentes salidas en cada ejecución lo que las hace **no determinísticas**. LangSmith permite realizar el **seguimiento completo del ciclo de vida** de la aplicación.



Traces



Un “**trace**” de LangSmith **agrupa todas las ejecuciones** (runs) generadas por una sola solicitud, funcionando como el ID que une todos los pasos de la operación (como llamadas a LLM).



Traces



Con el fin de gestionar el **volumen de datos generado** por los “**traces**” en LangSmith, los filtros permiten a los usuarios **identificar y examinar** rápidamente errores, fallos y cuellos de botella en el rendimiento de las aplicaciones LLM.

The screenshot shows the LangSmith interface for a project named 'chat-langchain-v3'. The 'Traces' tab is selected, and a red box highlights the 'Filters' section on the left. The filters include a 'Where' filter set to 'Is Trace' and an 'AI Query' filter. The main table displays a list of traces with columns for Output, Error, Start Time, and Latency. The right sidebar shows statistics for the project, including Run Count (8,311), Total Tokens (268,445,490), Median Tokens (19,387), Error Rate (1%), and % Streaming (100%).





02

Observabilidad en
Aplicaciones GenAI

Datasets



Los **Datasets** permiten realizar **evaluaciones repetibles y consistentes** en aplicaciones LLM. Estos conjuntos se componen de ejemplos que almacenan las entradas, las salidas y, opcionalmente, las salidas de referencia esperadas.

Runs	Threads	Monitor	Setup
1 filter	Last 7 days	Root Runs	LLM Calls
Name	Input	Output	Error
ChatOpenAI	human: will that work for heavy ai?	ai: I'm not sure if the provided steps will work specifically for Hea...	11/13/2024, 12:00
ChatOpenAI	human: Test the connection using a sample script to...	ai: ("tool_calls":[{"index":0,"id":"","call_5ZmFjL3CvV6SyoRz8wM...	11/13/2024, 12:00
ChatOpenAI	human: Check the LangChain documentation for any...	ai: ("tool_calls":[{"index":0,"id":"","call_IDABL0K48IML98VikDHS...	11/13/2024, 12:00
ChatOpenAI	human: Review the HeavyAI documentation to under...	ai: ("tool_calls":[{"index":0,"id":"","call_9JAA7UioG64ID8hFKFWLO...	11/13/2024, 12:00
ChatOpenAI	human: will that work for heavy ai?	ai: ("tool_calls":[{"index":0,"id":"","call_e4cCt5gMFv4zr3wFdB8Sc...	11/13/2024, 12:00
ChatOpenAI	human: RetrievalQAChain is deprecated	ai: Com a depreciação da classe 'RetrievalQAChain' no LangChai...	11/13/2024, 12:00
ChatOpenAI	human: Summarize the findings and provide a revise...	ai: ("tool_calls":[{"index":0,"id":"","call_DoLC7JqG0d4LT8Sib5Hz...	11/13/2024, 12:00
ChatOpenAI	human: Look for examples or guides specifically rela...	ai: ("tool_calls":[{"index":0,"id":"","call_kbXVIEj6H6gmf2nMIOx2hX...	11/13/2024, 12:00
ChatOpenAI	human: Check the latest LangChain documentation f...	ai: ("tool_calls":[{"index":0,"id":"","call_2bP2uVRpv5Z7SeVvngquZ...	11/13/2024, 12:00
ChatOpenAI	human: RetrievalQAChain is deprecated	ai: ("tool_calls":[{"index":0,"id":"","call_bIKoSLFM0fzRV24ngFKH2...	11/13/2024, 12:00
ChatOpenAI	human: E sobre a parte do rag?	ai: Para implementar um RAG (Retrieval Augmented Generation) c...	11/13/2024, 12:00
ChatOpenAI	human: Revisar a documentação sobre como config...	ai: ("tool_calls":[{"index":0,"id":"","call_LB08wBwPLGG0vKzzCKBL...	11/13/2024, 12:00
ChatOpenAI	human: Verificar exemplos de uso de RAG com mem...	ai: ("tool_calls":[{"index":0,"id":"","call_gWoZkDmLKQACURRUGiGJ...	11/13/2024, 12:00
ChatOpenAI	human: Consultar a documentação de integração do...	ai: ("tool_calls":[{"index":0,"id":"","call_YeGvQ3FXrtyNni98LSdaZ...	11/13/2024, 12:00
ChatOpenAI	human: E sobre a parte do rag?	ai: ("tool_calls":[{"index":0,"id":"","call_ErODfukN49GAqjIT13phnX...	11/13/2024, 12:00

Stats

Last 7 days

RUN COUNT

52,441

TOTAL TOKENS

254,293,661 / \$37.69

MEDIAN TOKENS

69

ERROR RATE

0%

% STREAMING

99%

LATENCY

P50: 115s P99: 28.9s

FIRST TOKEN

P50: 0.49s P99: 4.94s

Filter Shortcuts

Metadata

- LANGCHAIN_CALLBACKS_BACKGROUND
- LANGCHAIN_CALLBACKS_BACKGROUND == "true"
- LANGSMITH_AUTH_ENDPOINT
- LANGSMITH_AUTH_ENDPOINT == "https://api.smith..."
- LANGSMITH_AUTH_VERIFY_TENANT_ID
- LANGSMITH_AUTH_VERIFY_TENANT_ID == "true"
- LANGSMITH_FF_MULTIPART
- LANGSMITH_FF_MULTIPART == "true"
- LANGSMITH_HOST_PROJECT_ID

Selected 1 run

Add to Annotation Queue

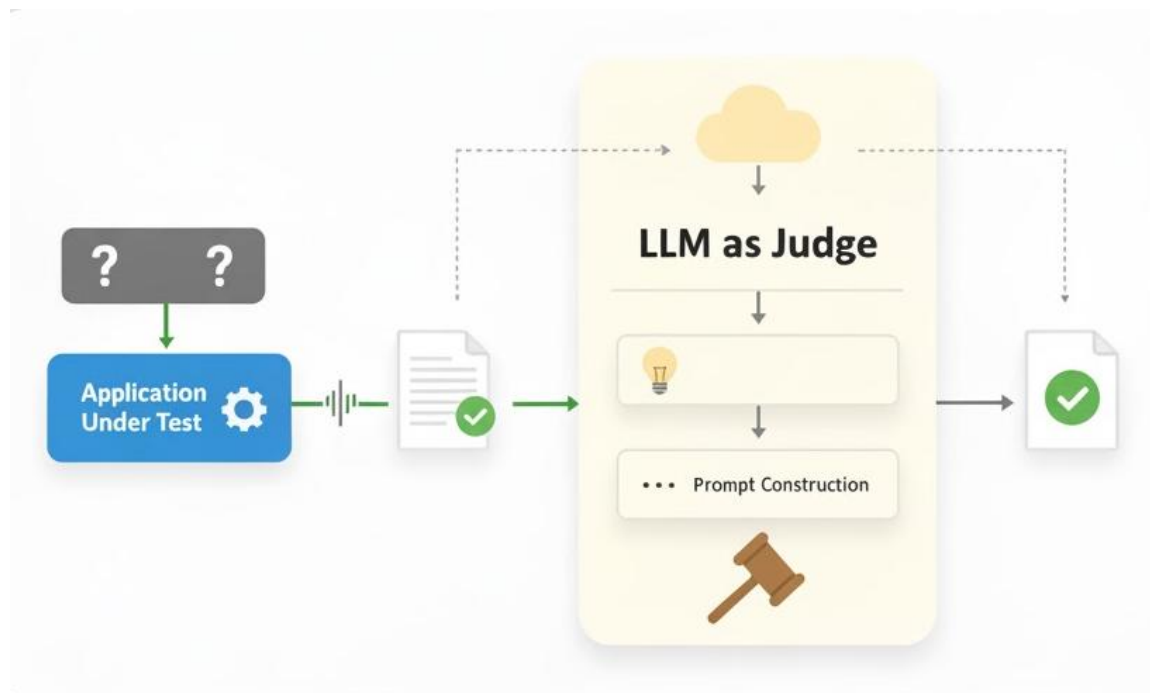
Add to Dataset

Cancel

Evaluadores



Los **Evaluadores son funciones** que califican el **rendimiento** de la aplicación LLM, utilizando un Ejemplo y la Ejecución (Run) generada para devolver métricas. Existen varias técnicas de evaluación, una de ellas es la de utilizar un **LLM como juez**.



Experimentos



Un Experimento es **el registro de resultados** que se obtiene al **probar** una versión específica de tu **aplicación** sobre un conjunto de datos para ver qué tan bien funciona.

LangChain Team > Datasets & Experiments > Email Extraction > Comparing 1 Experiments

Playground - 2025-02-04 03:56:04

Compact Full Diff JSON Heat Map Columns + Compare

Input	Reference Output	Output	Hallucination	Helpful	Random	Latency	Status	Tokens
OCTOBER 2023, VOL...	('#4eb6')	('tone': 'positive', 'topic': 'Princeton U... a: This email is a newsletter ... 1/3 < >	1.00	0.00 μ	0.50 μ	14.48s	SUCCESS 3	5,428
Solve the RedWorde We kn...	('#6a7d')	('tone': 'positive', 'topic': 'RedWorde ... a: The email is an engaging, ... 1/3 < >	1.00 μ	0.00 μ	0.50 μ	14.47s	SUCCESS 3	339,333
**Would you like to add spe...	('#72fb')	('tone': 'positive', 'topic': 'Personalize... a: This email is a promotion... 1/3 < >	1.00 μ	0.00 μ	0.50 μ	14.47s	SUCCESS 3	377,667
--- --- --- Book with Fal...	('#8602')	('tone': 'positive', 'topic': 'Fall Cruise ... a: This email is a promotion... 1/3 < >	1.00 μ	0.00 μ	0.50 μ	14.47s	SUCCESS 3	2,251,667
--- Costco --- ANSWER & ...	('#c4fa')	('tone': 'positive', 'topic': 'Invitation to... a: This email appears to be ... 1/3 < >	1.00 μ	0.33 μ	0.50 μ	12.98s	SUCCESS 3	578,333
USA Finance Solution offers ...	('#f1a8')	('tone': 'positive', 'topic': 'Business Fu... a: This email is a marketing ... 1/3 < >	1.00 μ	0.33 μ	0.50 μ	12.97s	SUCCESS 3	396
--- Adventure alongside ...	('#a7a2')	('tone': 'positive', 'topic': 'Pokemon G... a: This email is a promotion... 1/3 < >	1.00 μ	0.67 μ	0.50 μ	12.98s	SUCCESS 3	779,667
--- --- --- # We Prov...	('#b19a')	('tone': 'positive', 'topic': 'Financing o... a: This email appears to be ... 1/3 < >	1.00 μ	0.67 μ	0.50 μ	12.98s	SUCCESS 3	973,333
Hello Jacob, Do you know ...	('#c524')	('tone': 'positive', 'topic': 'Business Fl... a: This email is a promotion... 1/3 < >	1.00 μ	0.67 μ	0.50 μ	12.98s	SUCCESS 3	802,667
Hi Jacob, We are working to...	('#e419')	('tone': 'positive', 'topic': 'Feedback r... a: This email is a profession... 1/3 < >	0.67 μ	0.67 μ	0.50 μ	12.97s	SUCCESS 3	176
Happy Halloween, Jacob! Y...	('#e825')	('tone': 'positive', 'topic': 'Halloween ... a: This email is a festive, en... 1/3 < >	1.00 μ	0.67 μ	0.50 μ	12.97s	SUCCESS 3	776
Consider all ways to give to...	('#3864')	('tone': 'positive', 'topic': 'Donation an... a: The email is a heartfelt ap... 1/3 < >	1.00 μ	1.00 μ	0.50 μ	14.48s	SUCCESS 3	437,333
Some travelers plan ahead...	('#486d')	('tone': 'positive', 'topic': 'Travel and ... a: This email is a promotion... 1/3 < >	1.00 μ	1.00 μ	0.52 μ	14.48s	SUCCESS 3	384,333
Hello Jacob, For many small...	('#55cf')	('tone': 'positive', 'topic': 'Business fl... a: This email is a marketing ... 1/3 < >	1.00 μ	1.00 μ	0.50 μ	14.48s	SUCCESS 3	585,667
### Where sustainability ...	('#8412')	('tone': 'positive', 'topic': 'Promotion ... a: This email is a promotion... 1/3 < >	1.00 μ	1.00 μ	0.50 μ	14.47s	SUCCESS 3	308





03

Gestión de prompts

Gestión de Prompts



LangSmith ofrece un sistema para gestionar y versionar prompts. La plataforma proporciona control de versiones mediante commits, facilitando la iteración y prueba en el prompt playground y sobre conjuntos de datos.

The screenshot displays the LangSmith interface for managing prompts. On the left, a list of commits for the prompt 'email-assistant' is shown, including commit IDs, timestamps, and counts of messages and evaluations. The right panel shows the details of the selected commit (84819a32), including the system prompt and response categories.

email-assistant Private [Edit] [Delete] [Make Public] [Fork] [Try it]

Updated a few seconds ago • 0 hearts • 20 views • 0 downloads

Prompt **Commits** 6

84819a32 a few seconds ago 2 messages 0 evaluations

fd6e44bf a minute ago 1 message 0 evaluations

f7b34edb an hour ago 5 messages 0 evaluations

09f20dfa 16 hours ago 6 messages 0 evaluations

ff0a688e 16 hours ago 2 messages 0 evaluations

6b69c03b 16 hours ago 4 messages 0 evaluations

84819a32 Messages Model Configuration Show diff

SYSTEM
You serve as the executive assistant. Your primary responsibility is determining if the following email requires a direct response.

Response Categories:

- 'email' - Requires a direct response (HIGH PRIORITY):
 - Direct business opportunities or partnership discussions
 - **First-time introductions needing personal acknowledgment**
 - Critical business decisions requiring CEO approval
 - Direct questions about the company's strategy or vision
 - **Time-sensitive scheduling requests requiring his confirmation**
- 'notify' - Important but no response needed (MEDIUM PRIORITY):
 - FYI updates on ongoing projects
 - Conference/event invitations without immediate deadlines
 - Calendar updates that don't conflict with existing commitments
 - General product feedback or suggestions
 - Industry news or updates
 - **First-time introductions needing personal acknowledgment**
- 'no' - No action required (LOW PRIORITY):
 - Automated notifications or system updates
 - Marketing emails or newsletters
 - Calendar acceptances/confirmations
 - Thread updates where the user isn't primary recipient
 - General announcements without specific requests



Notebook: Prompts



Participación

1. Elijan un caso de uso en el que justifiquen la utilización de la observabilidad y las evaluaciones.
2. Tendrán 20 minutos para crear la presentación y 5 minutos para exponer.



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**II PROGRAMA DE ESPECIALIZACIÓN EN IA
GENERATIVA Y MACHINE LEARNING OPS**

Gracias